

Anti-fouling amphiphilic surfaces from norbornene-based fluorinated copolymers with excellent mechanical properties

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ABSTRACT

Amphiphilic coatings have garnered significant attention in the realm of fouling resistance due to their potential to reduce fouling adhesion and enhance fouling release. However, developing anti-fouling amphiphilic surfaces with robust mechanical properties remains a formidable challenge. Herein, a series of norbornene-based amphiphilic copolymers (PNPFE-PEG-*x*) with varying NPEG mass fractions (0–50 wt%) were well-designed and fabricated via living ring-opening metathesis polymerization (ROMP) employing norbornene-based fluorinated monomer (NPFE) and poly(ethylene glycol) (NPEG) monomer. The water absorption, surface wettability, and mechanical properties of the amphiphilic copolymers can be regulated by adjusting the NPEG contents. Particularly, dry PNPFE-PEG-10 % films exhibited exceptional tensile strength of 14.36 MPa and breaking elongation of 376 %. The wet PNPFE-PEG-H₂O samples exhibited a unique hydration-induced stiffening phenomenon. Even in the water swelling state, wet PNPFE-PEG-50 % films still demonstrated remarkable tensile strength of 8.05 MPa and breaking elongation of 74.81 %. Furthermore, increasing the NPEG content effectively resisted bovine serum albumin and fibrinogen protein adsorption on the coatings, with PNPFE-PEG-50 % showing the lowest protein adsorption at 19 µg/cm² and 27 µg/cm², respectively, indicating excellent anti-fouling properties. In addition, PNPFE-PEG-*x* exhibited anti-icing property through rational molecular design. These high-performance anti-fouling coatings are equipped to meet diverse application needs in marine anti-fouling, biomedical instruments, and other relevant fields.

1. Introduction

The adsorption of microorganisms such as proteins and bacteria on surfaces can have serious consequences in various applications, including marine facilities and biomedical instruments [1,2]. Anti-protein adhesion is extremely important for anti-fouling coatings as protein is firstly absorbed by substrate materials via molecular interactions to form a conditioning layer among the several key stages in the attachment of fouling organisms [3]. This has led to an increasing need for the development of anti-protein or anti-fouling materials. These materials can be divided into three types of anti-fouling coating surfaces: hydrophobic [4], hydrophilic [5], and amphiphilic fouling resistance materials [6]. Hydrophobic anti-protein adsorption materials, such as silicone and fluorine-based fouling release coatings, have garnered significant attention due to their low surface energy and weak adhesion to

microorganisms [7–9]. However, these coatings often exhibit poor adhesion strength to the substrate, as the anti-fouling ability conflicts with substrate adhesion. On the other hand, hydrophilic anti-protein adsorption materials, like poly (ethylene glycol) (PEG) and zwitter-ionic polymers, aim to prevent protein adhesion through the formation of a hydration layer or space obstruction [10,11]. However, they typically demonstrate poor substrate adhesion and mechanical properties, greatly limiting their scope of applications [12].

As a solution, an integrated design of hydrophilic and hydrophobic components into anti-fouling materials with simultaneous mechanical strength and anti-fouling adsorption is required to meet the requirements of various applications. For instance, a kind of amphiphilic interpenetrating polymer network (IPN) has been constructed, in which the hydrophobic silicon-containing epoxy resin imparts superior mechanical properties and excellent adhesion, and the hydrophilic

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hydrogel part endows surface with excellent anti-fouling properties [13]. In addition, a high-performance anti-fouling coating was fabricated by introducing 2-ureido-4[1H]-pyrimidinone (UPy) and amphiphilic pendant chains to a silicone-based polyurethane, presenting great adhesion strength and good fouling anti-fouling performance [14]. Therefore, well-designed amphiphilic materials with tailored hydrophilic and hydrophobic compositions regulating mechanical strength and anti-protein adsorption is vital of importance [15]. To achieve this goal, various synthetic methodologies have been developed, such as conventional free radical polymerization, post-modification of backbones, and condensation polymerization, but they are lack of well-controlled structures [16–18]. In recent years, researchers have utilized atom transfer radical polymerization (ATRP) to obtain amphiphilic polymers with well-controlled structures [19,20]. However, few attempts have been made to reveal the effect of different functions on mechanical strength and fouling resistance property of amphiphilic polymers.

Ring-opening metathesis polymerization (ROMP) is a feasible and highly efficient synthetic approach, which allows for the synthesis of well-designed amphiphilic polymers with controlled molecular compositions at mild conditions and fast polymerization rates [21]. According to our previous study, polynorbornene can be produced via living ROMP and has been shown to yield materials with high strength and high transparency, making it suitable for the backbones of amphiphilic polymers [22].

Herein, based on the low surface energy of fluorine-containing polymers, the protein-resistant property of polyethylene glycol (PEG), and the high strength of polynorbornene and rigid pentafluorophenyl side chain, a series of amphiphilic copolymers (PNPFE-PEG-*x*) with adjustable compositions are facilely prepared through ROMP of norbornene-based fluorinated monomer (NPFE) and PEG monomer (NPEG) in various mass ratios, as shown in Scheme 1. The effects of molecular compositions on the surface morphology, surface wettability, mechanical strength, and anti-fouling properties have been comprehensively studied, as evidenced from AFM and tensile experiments, etc. Benefiting from the well-designed amphiphilic polymers, hydration layers formed on the amphiphilic coating surfaces by binding with water molecules of NPEG segment and low surface energy of PNPFE segment, contributing to the excellent anti-protein properties. This work will indicate the cooperative effect of the hydrophobic and hydrophilic chain segments on mechanical strength and fouling resistance, contributing to understanding the structure-property relations and design of materials from the molecular control level.

2. Materials and methods

2.1. Materials

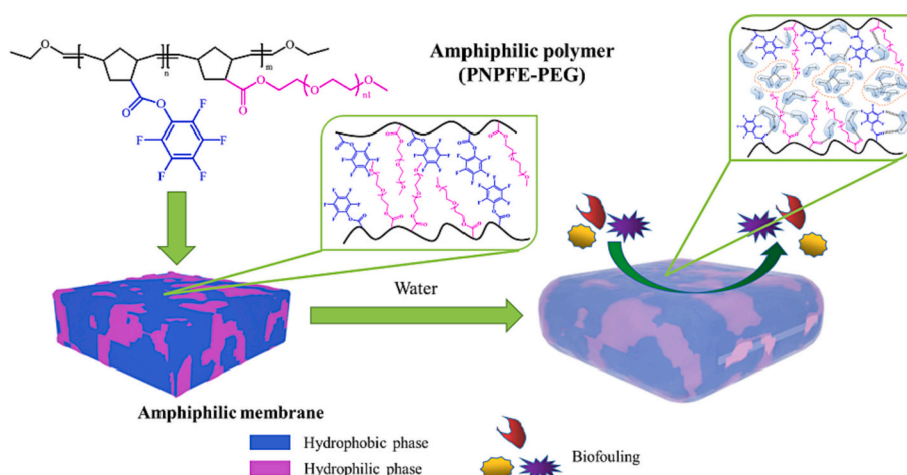
5-Norbornene-2-carboxylic acid (NCA, 99 %), methoxypolyethylene glycols (MPEG, >98.0 %, 550 g/mol), 2,3,4,5,6-pentafluorophenol (PFP, 99 %), 4-dimethylaminopyridine (DMAP), ethyl vinyl ether (EVE, 98 %), trimethylacetic anhydride (TA, >98 %), N-(3-(dimethylamino)propyl)-N'-ethylcarbodiimide hydrochloride (EDCI, 98.0 %), Grubbs 2nd-generation catalyst (98 %) were used directly without further purification. Bovine serum albumin (BSA) were used as received from Shanghai Maclean Biochemical Technology Co., Ltd. Fibrinogen (Fg) was purchased from Beijing Solaibao Technology Co., LTD. Phosphate buffered solution (PBS; 0.1 mol·L⁻¹, pH 7.4) were obtained from Beijing Solarbio Science & Technology Co., Ltd. Isothiocyanate fluorescent marker bovine serum albumin (FITC-BSA) was purchased from Shanghai Xinfan Science & Technology Co., Ltd. Methylene chloride, tetrahydrofuran and other solvents were used directly without further purification.

2.2. Synthesis of norbornene-based fluorinated monomer (NPFE)

NCA (1.38 g, 10 mmol) and PFP (2.02 g, 11 mmol, 1.1 equiv) dissolved in 50 mL methylene chloride were added into a 100 mL three-necked flask. Then EDCI and DMAP were added into the flask at 0 °C. The reaction was carried out at room temperature for 24 h under N₂ protection. The resulting product was poured into saturated ammonium chloride solution, extracted by ethyl acetate, dried with anhydrous magnesium sulfate. The organic phase was collected and concentrated. The pure NPFE product (2.8 g, 9.5 mmol, yield: 82.39 %) was obtained after purification by column chromatography (n-hexane/ethyl acetate =96:4).

2.3. Synthesis of norbornene-based PEG monomer (NPEG)

NCA (1.38 g, 16 mmol), MPEG (7.70 g, 14 mmol) and TA (2.98 g, 16 mmol) dissolved into 50 mL THF were added into a 100 mL three-necked flask. Then DMAP (17.10 mg, 0.14 mmol) was added into the flask. The reaction was carried out at 60 °C for 24 h under N₂ protection. The resulting product was successively washed by saturated NaHCO₃ solution, NaCl solution and distilled water, extracted by CH₂Cl₂, dried with anhydrous magnesium sulfate. The organic phase was collected and concentrated. Viscous oil-like liquid NPEG (7.65 g, 11.2 mmol yield: 85.42 %) was obtained.



Scheme 1. Schematic of surface antifouling of amphiphilic polymer (PNPFE-PEG-*x*).

2.4. Synthesis of norbornene-based polymer (eg: PNPFE-PEG-50 %)

NPEG (2.68 g, 4 mmol) and NPFE (1.22 g, 4 mmol) dissolved in methylene chloride (DCM) (30 mL) were added into a 50 mL three-necked flask. Then the Grubbs 2nd catalyst (0.04 g) was added to the flask at 0 °C. The reaction was carried out at 30 °C for 12 h under N₂ protection and terminated by adding a small amount of ethyl vinyl ether (EVE) (5 mg). The resulting product was concentrated and washed with anhydrous ethanol to remove catalyst and unreacted monomers. Finally, light brown solid of amphiphilic polymers PNPFE-PEG-50 % was obtained. Using the above polymerization steps, other proportions of polymers are polymerized by adding corresponding proportions of monomers.

2.5. Preparation of amphiphilic polymer films

The resulting amphiphilic polymers were dissolved in tetrahydrofuran with a concentration of 30 wt%. The polymer solutions were poured into the PTFE molds. After evaporation at room temperature, the films were then annealed at 60 °C in a vacuum oven for 12 h. Finally, smooth and uniform films were obtained with the thickness of 200–300 μm.

As for the coatability, the amphiphilic polymer of PNPFE-PEG-30 % is fully dissolved in tetrahydrofuran with a concentration of 37 wt%. The polymer solution is evenly coated on the polytetrafluoroethylene, glass and aluminum substrates.

2.6. Characterization

¹H NMR analysis was recorded on a Bruker AV400 MHz spectrometer by dissolving the samples into a suitable deuterated solvent at room temperature. **¹³C NMR analysis** was recorded on a Bruker AV400 MHz spectrometer by dissolving the samples into a suitable deuterated solvent at room temperature. The **ATR-FTIR** was recorded on a Bio-Rad FTS-135 spectrometer in the scan range of 4000–500 cm⁻¹. Number-average molecular weight (M_n), weight-average molecular weight (M_w) and polydispersity index (Đ) of polymers are determined by **Gel Permeation Chromatography** (GPC) (Waters 1500, USA) equipped with two separation columns (Waters, KF-804 and KF-805 columns; eluent: THF; flow rate: 1.0 mL/min, at 35 °C) calibrated with polystyrene (PS) standards, and one 2414 differential RI detector. **Atomic force microscope** (AFM) measurements were performed using a multi-mode AFM machine (CSPM5500A of Ben Yuan Ltd., China) in the tapping mode at room temperature. The **static water contact angles** were measured by an optical contact angle meter system (Data-physics OCA40, Germany). Drops were fitted with a Young-Laplace formula to calculate the static contact angle in the Theta software (Biolin Scientific). Each data presented was the average values from more than five measurements. **Polarized optical microscope** was collected using the Nikon LV100POL optical microscope at 20× magnification. **DSC** were measured by a Setline DSC differential scanning calorimeter with a nitrogen flow rate of 10 mL/min and a heating rate of 5 °C/min within the temperature range of 10–80 °C to obtain the corresponding heat flow curve.

2.7. Water uptake test

The surface properties, anti-fouling and anti-icing properties of the amphiphilic coatings are associated with their hydration layer formed by NPEG segment with water molecules through H-bonds, which was strongly influenced by the water uptake rate (S%) of the amphiphilic polymers. Herein, the weighing method was used to measure the amount of water absorbed by amphiphilic polymers. Dried films were weighed and recorded as m₀, which were immediately submerged in deionized water for 24 h. After the second immersion cycle, the wet films were removed and lightly patted with a Kimwipe to remove free

surface water and subsequently weighed as m₁. Water absorption (S%) was calculated with the formula shown below: $S\% = (m_1 - m_0) / m_0 \times 100\%$.

2.8. Tensile test

The mechanical properties were determined by a tensile test using an Instron 5569 machine at a speed of 200 mm/min and a load cell of 50 N. The original size of the film specimens was measured using a Vernier caliper to calculate the stress and strain at a given load.

2.9. Anti-protein test

Isothiocyanate fluorescent marker bovine serum albumin (FITC-BSA) was diluted with PBS buffer to a concentration of 0.1 mg mL⁻¹. The amphiphilic films were immersed in the as-prepared FITC-BSA solution placed on a rocking platform at 37 °C in the dark at the speed of 100 rpm. After 2 h, the samples were removed from the solution and washed five times with 1 mL PBS buffer. The prepared samples were dried with N₂, and were then observed on a confocal laser scanning microscope (Carl Zeiss, Germany). The FITC-BSA adsorption on the samples was observed at λ_{ex} = 488 nm/λ_{em} = 520 nm for detection of the FITC dye.

To quantitatively test the protein adsorption on the coating surfaces, all films (length*width = 1 cm*1 cm) were pre-wetted in PBS (pH = 7.4) buffer solution, followed by BSA or fibrinogen (Fg) solution incubating for 12 h at room temperature. After being washed 5 times with PBS buffer solution, the samples were placed into 2 mL PBS buffer solution and then performed a ultrasound at room temperature for 20 min to remove the protein adsorbed on the surface. According to the instructions of the BCA kit, the protein elution solution of each sample was added in centrifuge tubes containing BCA reagent at 37 °C and remained for 30 min. The amount of the adsorbed protein was determined by the adsorption intensity at 562 nm using UV-Vis spectrophotometer (TU-1901, China). The calibration curve was pre-set by measuring different protein concentrations in PBS buffer solution. The result data was the mean value of five samples for each sample.

2.10. Anti-freezing test

A semiconductor thermostat was used to measure the ice nucleus temperature on the amphiphilic polymer surfaces, which was performed to evaluate the anti-icing performance. The samples were placed on a 0 °C cooling table, and 10 μL deionized water was added to the surface and kept for 2 h. Then the temperature was cooled down at a rate of 1 °C/min. The samples were maintained for 2 h every 1 °C decrease in temperature until the temperature that the water droplet changed from transparent to opaque was defined as the heterogeneous ice nucleation temperature.

3. Results and discussion

3.1. Synthesis and characterization of amphiphilic polymers

The detailed synthesis of PNPFE-PEG-x is shown in Scheme S1. Initially, hydrophilic and hydrophobic monomers (NPEG and NPFE) were prepared by 5-norbornene-2-carboxylic acid (NCA) with poly(ethylene glycol) (PEG) and 2,3,4,5,6-pentafluorophenol (PFP), respectively. The confirmation of the two monomers was carried out through analysis using ¹H and ¹³C NMR spectroscopy as well as FT-IR spectroscopy, as illustrated in Figs. S1 and S2. Subsequently, PNPFE-PEG-x bearing varying NPEG contents was facilely implemented through ROMP by mixing NPEG and NPFE in different mass ratios in the presence of Grubbs 2nd catalyst. As shown in Fig. 1a, the structures for the polymers were verified by ¹H NMR, which was confirmed by the peak shift of double bonds from δ (ppm) = 5.95–6.35 of NPEG and NPFE to δ (ppm) = 5.20–5.65 (-CH=CH- in the amphiphilic polymeric

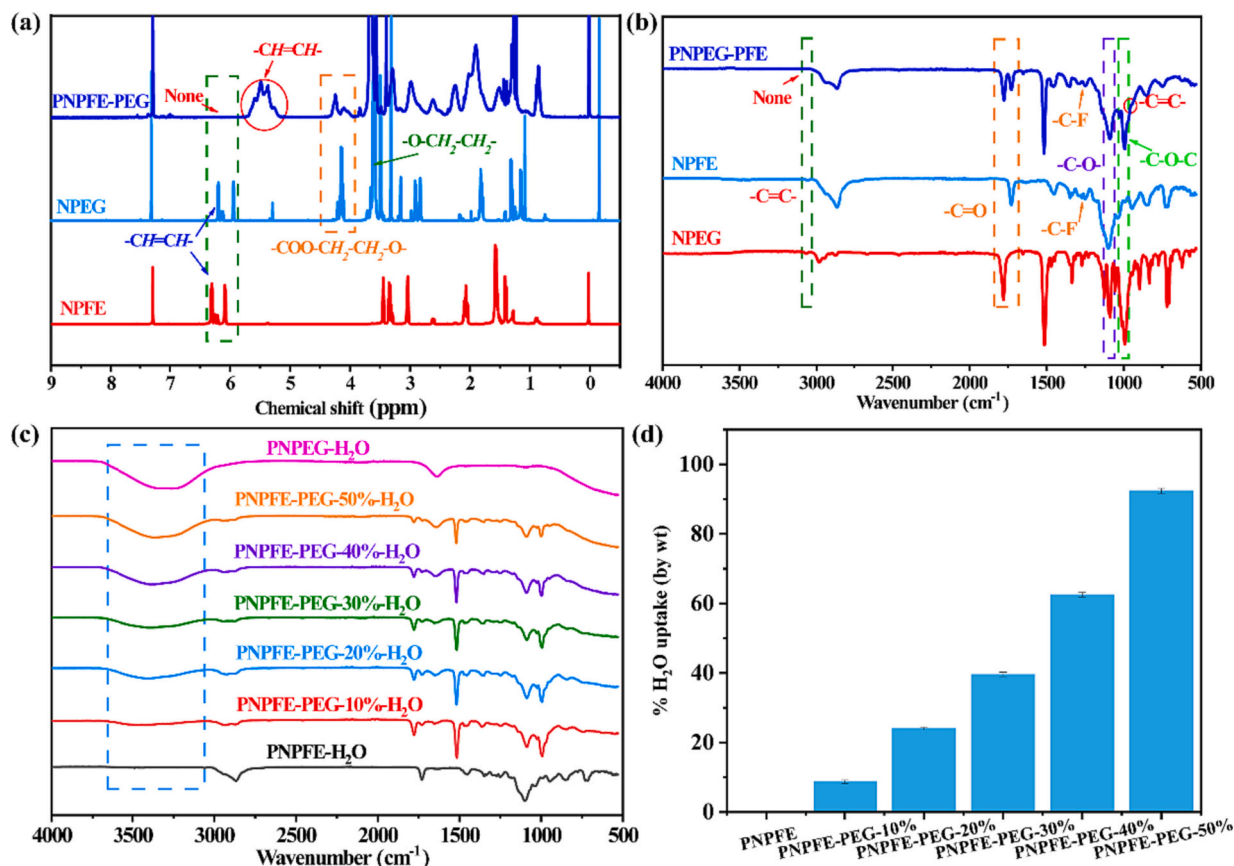


Fig. 1. ^1H NMR spectra (a) and FT-IR spectra (b) of NPFE, NPEG, and PNPF-PEG, (c) FT-IR spectra of water swelling films and Wt% H₂O uptake of the PNPF-PEG-x (d).

backbones), and the occurrence of the peaks at δ (ppm) = 3.58 ($-\text{O}-\text{CH}_2-\text{CH}_2-$) and δ (ppm) = 4.16 ($-\text{COOCH}_2\text{CH}_2\text{O}-$) of NPEG. When NPEG contents increased from 0 to 50 wt%, the characteristic peaks at δ (ppm) = 4.16 of NPEG increased in intensity (Fig. S3). Furthermore, the as-prepared polymers and monomers were characterized by ATR-FTIR, as shown in Fig. 1b. The characteristic bands appeared at 1729, 1783 cm^{-1} are assigned to $\text{O}=\text{C}$ of NPEG and NPFE, the characteristic peaks of $\text{C}-\text{O}$ of in ester groups in NPFE and NPEG are 1100, 1088 cm^{-1} , respectively. And the peak at 990 cm^{-1} is assigned to $\text{C}-\text{O}-\text{C}$ of NPEG. Furthermore, the disappearance of characteristic absorption peaks at 3060 cm^{-1} of double bonds from the norbornene ring in NPEG and NPFE and the appearance of a new peak at 960 cm^{-1} assigned to the double bonds in the polymeric backbones, in addition, the appearance of the peak at 1230 cm^{-1} ascribed to $\text{C}-\text{F}$ bond from NPFE in FT-IR, indicating the successful preparation of amphiphilic polymers with varying NPEG contents (Table S1). GPC curves show a symmetric monomodal distribution ($\text{Đ} = 1.51\text{--}1.63$) and well-controlled number-average molecular weight ($M_n = 119.1\text{--}139.2$ kg/mol) through precise control of the mole ratio of monomers to catalyst, as shown in Fig. S4 and summarized in Table S2.

3.2. Water swelling properties

Since the hydrophilic NPEG chain segments could bind water molecules through the formation of H-bonding interactions, ATR-FTIR experiments were carried out to reveal the H-bonds by immersing the films in deionized water for 24 h. As shown in Fig. 1c, the absence of significant peak shifts in the PNPF-PEG-H₂O spectrum compared to that of dry PNPF films suggests the stability of PNPF films underwater. Upon the introduction of hydrophilic NPEG chain segments, a new broad peak appeared at 3050–3680 cm^{-1} in water swelling amphiphilic films,

which can be divided into two peaks at 3250 and 3400 cm^{-1} . According to the literature [23], the peak observed at 3250 cm^{-1} is attributed to the formation of hydrogen bonds between the hydrophilic NPEG segments and water molecules, representative of non-freezable water. The peak at 3400 cm^{-1} , on the other hand, corresponds to the stretching of residual water, signifying freezable water. Therefore, the appearance of this new broad peak is ascribed to the presence of both the hydrogen bonding band and stretching of residual water in comparison with the dry films. The intensity of the peak significantly strengthened with the increase of NPEG contents, demonstrating enhanced water swelling properties [24–26]. Consequently, hydration layers can form on the amphiphilic coating surfaces by binding with water molecules, which is beneficial for resisting fouling adsorption and inhibiting ice formation in extreme environments.

Additionally, water absorption properties of the amphiphilic polymers were evaluated by immersing the films in deionized water for 24 h. As shown in Fig. 1d, an increase in water uptake rate from 0 to 90 wt% was observed with an increase in NPEG content from 0 to 50 wt%, which is in good agreement with that the H-bonding interactions enhanced as increasing the NPEG contents, as confirmed by ATR-FTIR. This allowed more water molecules into the bulk of the films as the NPEG content increased, thereby generating a water-swelling network [27,28]. Moreover, polarizing microscope observations revealed that the aggregation domain of bound water significantly increased with an increase in NPEG content, as illustrated in Fig. S5. Therefore, we assessed the mechanical properties and anti-fouling properties of the amphiphilic copolymers in both dry and water-swelling state to evaluate their potential application in different environments.

3.3. Surface morphology

AFM experiments are performed to observe the 3D surface topography of dry and water swelling amphiphilic coatings. Initially, the surface of pure PNFE coating in dry state appeared smooth with a small roughness of 0.59 nm. Upon introducing hydrophilic NPEG, significant phase separation morphology features were observed in all the coatings due to the incompatibility of the hydrophobic PNFE segments and hydrophilic NPEG segments (Fig. 2) [29,30]. Furthermore, the surface roughness (Ra) initially decreased and then increased with an increase in hydrophilic NPEG content, following the order PNPFE-PEG-10 % (Ra = 1.07 nm) > PNPFE-PEG-20 % (Ra = 0.95 nm) > PNPFE-PEG-30 % (Ra = 0.61 nm) < PNPFE-PEG-40 % (Ra = 1.31 nm) < PNPFE-PEG-50 % (Ra = 3.41 nm). Nanometer morphology features were distributed on the surface of all formulations, which are similar to those previously reported by Tiller and Wooley [31,32]. The surface topography of the films underwent significant changes after swelling in water due to the dynamic recombination of surface features. Regarding the samples with the same NPEG content, the surface roughness of the water swelling samples was notably higher than that of the dry samples, indicating the hydration-driven microphase separation. When the NPEG content reached 50 wt%, the surface roughness of PNPFE-PEG-50 % increased to 16.90 nm. Significantly, the coating surfaces of both dry and wet samples exhibited nano-scale structures, which serve as terrain barriers for the accumulation of ice and the settlement of microorganisms [33,34].

3.4. Surface wettability

The surface wettability of the amphiphilic coating material was analyzed by determining the static water contact angle (WCA) on the polymer surfaces. Fig. 3a shows that the WCA on the dry PNPFE surfaces is 90.8° due to hydrophobic nature of fluorinated polynorbornene. The WCA gradually decreased as the NPEG content increased for dry samples, because the increase in amount of surface hydrophilic NPEG effectively increased the surface hydrophilicity, which was verified by the energy dispersive spectrometer (EDS) result that the proportion of oxygen element on the surfaces gradually increased and the proportion of fluorine element decreased as NPEG content increased, as shown in Fig. S6.

To further understand the surface hydration and underwater restructuring of the amphiphilic polymers, we recorded the time-dependent changes in water contact angles on the dry films. Fig. 3b shows that the contact angle on the PNPFE homopolymer surface was 90.8° and remained unchanged over time due to the surface stability. In contrast, the initial contact angles of the PNPFE-PEG-x surfaces exhibited a significant decrease over time. As the water droplets spread over

time, the surfaces underwent a reconstruction that the NPEG segment migrated to the interface between water and coating, thereby leading to an increase of hydrophilicity. Moreover, the reduction rate of the contact angles increased as the hydrophilic NPEG contents increased. Notably, the water contact angle on the amphiphilic surfaces decreased to 64.9°, 60.1°, 46.2°, 36.8°, and 22.1° after 300 s, as the NPEG contents were 10 %, 20 %, 30 %, 40 % and 50 %, respectively. Due to the prominent hydrophilicity of PNPEG homopolymer, water droplets could quickly spread within 125 s and the water contact angle on the PNPEG surface decreased to 0°. Overall, the amphiphilic surface properties changed after immersion in water due to the surface reconstruction and the formation of water layer [35,36], which was beneficial for the protein resistance and icing inhibition.

3.5. Mechanical properties

Generally, excellent amphiphilic anti-fouling materials require mechanical robustness or toughness, both in dry state and aqueous environment to meet the practical applications [37]. The mechanical properties of the polymer films in dry and water swelling states were evaluated using stress-strain measurements at a strain rate of 200 mm·min⁻¹ at room temperature, as depicted in Fig. 4 and listed in Table S3. The dry PNPFE homopolymer films exhibited brittle fracture characteristics in the stress-strain curves, displaying high tensile strength of 10.98 MPa, mainly due to the rigid fluorinated polynorbornene chain segment conferring high strength and restricting chain mobility. The incorporation of NPEG chain segments significantly altered the mechanical properties of the amphiphilic polymers. Increasing the NPEG content from 10 to 30 wt% led to flexible fracture in dry PNPFE-PEG-x films, with high tensile stresses of 14.36, 13.07, and 8.73 MPa and improved breaking elongations of 217, 249, and 300 %, respectively. While exceeding 30 % NPEG content rendered the polymer films too soft to be peeled off in a dry state. The mechanical properties of PNPFE-PEG-40 % and PNPFE-PEG-50 % were not able to be determined. As shown in Fig. S7, the DSC results showed that the glass transition temperatures (T_g) consistently decreased with the increase of NPEG contents. Notably, it was difficult to identify the T_g from the DSC curve of PNPFE-PEG-50 % at temperatures exceeding 15 °C. The results indicated that NPEG could concurrently increase molecular chain mobility and chain entanglement. When the NPEG content is lower than 20 wt%, the rigid PNPFE and flexible NPEG chain segments, as well as chain entanglement between PNPFE and NPEG segments cooperatively improved the mechanical strength and breaking elongations compared to the PNPFE films. The introduction of more NPEG contents endowed the materials with high chain mobility while sacrifice the mechanical strength. Consequently, the tensile stress initially increased and then

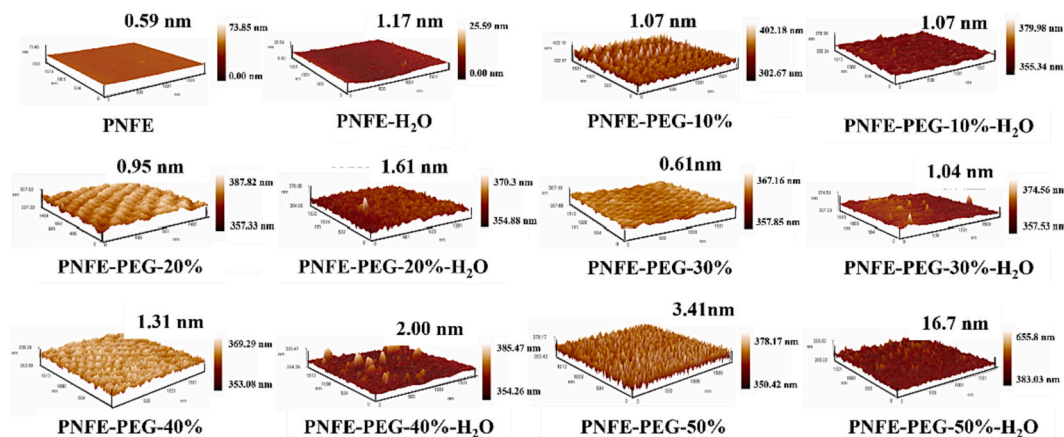


Fig. 2. Atomic force microscopy images for the amphiphilic coatings (PNPFE-PEG-x) array in both the dry (left) and water-swollen (right) states. Scan size is $2 \times 2 \mu\text{m}^2$.

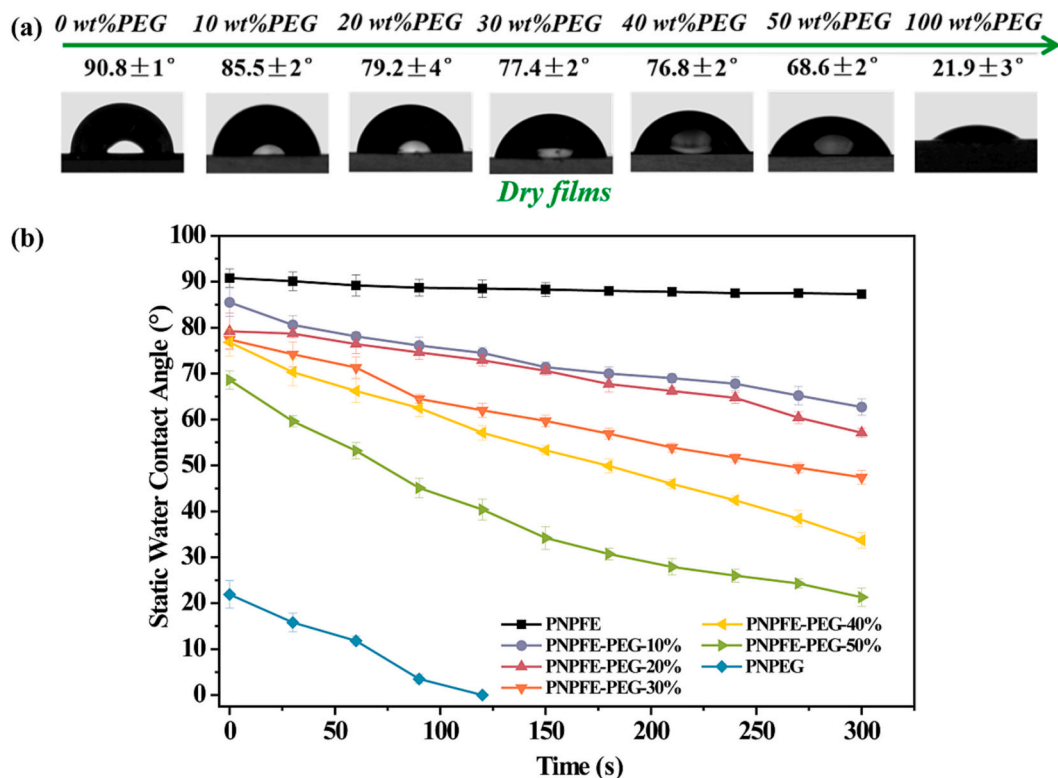


Fig. 3. Static water contact angles (a) of the dry amphiphilic coatings (PNPFE-PEG-x) in the and time dependence of the water contact angle (b).

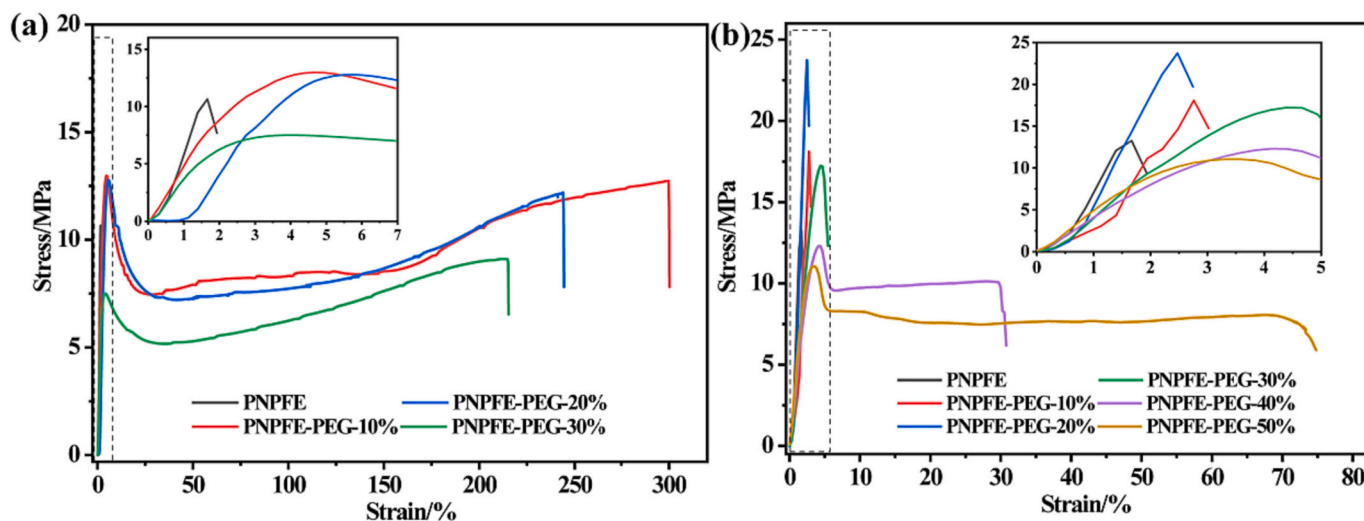


Fig. 4. The stress-strain curves of the amphiphilic coatings (PNPFE-PEG-x) in both the dry films (a) and water swelling films(b).

decreased, with PNPFE-PEG-x films containing NPEG contents higher than 40 wt% displaying an extremely soft nature.

Further, amphiphilic polymers in wet states exhibited distinct mechanical properties after being immersed in water for 24 h. The wet PNPFE-PEG-H₂O samples exhibited a unique phenomenon of hydration-induced stiffening, presenting brittle and hard characteristics in contrast to the dry films with the same NPEG content. If the NPEG content was lower than 30 wt%, the wet films exhibited brittle fracture characteristics, contrasting with the elastic features of the dry films, and showed extremely high tensile strength ranging from 13.26 to 23.72 MPa. The tensile stress of PNPFE-PEG-10 %, PNPFE-PEG-20 %, and PNPFE-PEG-30 % significantly increased by 1.26, 1.81, and 1.97 times, respectively,

compared to the stress in the dry state. Surprisingly, the extremely soft PNPFE-PEG-40 % and PNPFE-PEG-50 % films, when soaked in water, exhibited elastic characteristics with high stresses of 10.12 and 8.05 MPa, and elongations at break of 30.8 % and 74.8 %, respectively, in the PNPFE-PEG-40 %-H₂O and PNPFE-PEG-50 %-H₂O samples. These results were primarily attributed to the hydration-driven microphase separation, as confirmed by the AFM results. In addition, NPEG crystallization might also cause the stiffening of the amphiphilic PNPFE-PEG-x upon hydration [38,39]. In comparison with previous findings, the well-designed norbornene-based amphiphilic copolymers exhibited exceptional mechanical strength (>8.73 MPa) in both water and air environments, significantly surpassing the tensile strengths of the

reported PEG-based amphiphilic coatings (approximately 3 MPa) [10,11,40]. These results strongly indicated that the composition of the hydrophobic/hydrophilic chain segments could effectively tune the mechanical properties to meet the requirements of various applications.

3.6. Protein resistance

Before assessing the protein resistance of the amphiphilic polymers, their ability to coat on suitable substrates was examined. The amphiphilic copolymer PNPFE-PEG-30 % was applied to various substrates such as polytetrafluoroethylene, glass, and aluminum plates, as depicted in Fig. S8. It was evident that the polymers adhered well to all the aforementioned substrates. Anti-protein adhesion is the crucial parameter for anti-fouling coatings as protein is firstly absorbed by substrate materials via molecular interactions to form a conditioning layer among the several key stages in the attachment of fouling organisms [13]. As shown in Fig. 5a, fluorescently labeled bovine serum albumin (FITC-BSA) served as a model to test the anti-protein adhesion properties, characterized by confocal laser scanning microscopy. The Image J software was utilized to statistically analyze the fluorescence intensity (Fig. 5b), with the control PNPFE homopolymer normalized as 100 %.

Obvious green fluorescence was observed on the PNPFE surface, indicating an amount of protein adsorption. As the NPEG content increased from 10 wt% to 50 wt%, the fluorescence brightness of the amphiphilic surface gradually decreased, with fluorescence intensities of 85.94 %, 51.08 %, 43.03 %, 36.72 %, and 19.77 %, respectively, indicating an enhanced effect against protein adhesion. Particularly, little fluorescence with extremely low intensity was visible on the PNPFE-PEG-50 % surface, demonstrating excellent anti-fouling properties.

Furthermore, the anti-fouling durability of the surface was studied by FITC-BSA adsorption on the polymer surface for 3 cycles. As depicted in Fig. 5b, the fluorescence intensity slightly increased with an increasing number of cycles for the control PNPFE and amphiphilic polymers with NPEG content <20 %. However, as the NPEG content increased further, there was no remarkable change in the fluorescence intensity of PNPFE-PEG, confirming the good durability of the anti-fouling coating. The result can be attributed to the formation of a hydration layer on the surface by hydrophilic NPEG, which generated hydrogen bonding interactions with water molecules, thus inhibiting protein adhesion [41–43]. Additionally, the highly ordered and complex surface morphology at the nanometer scale as confirmed by AFM creates resistance to protein adsorption.

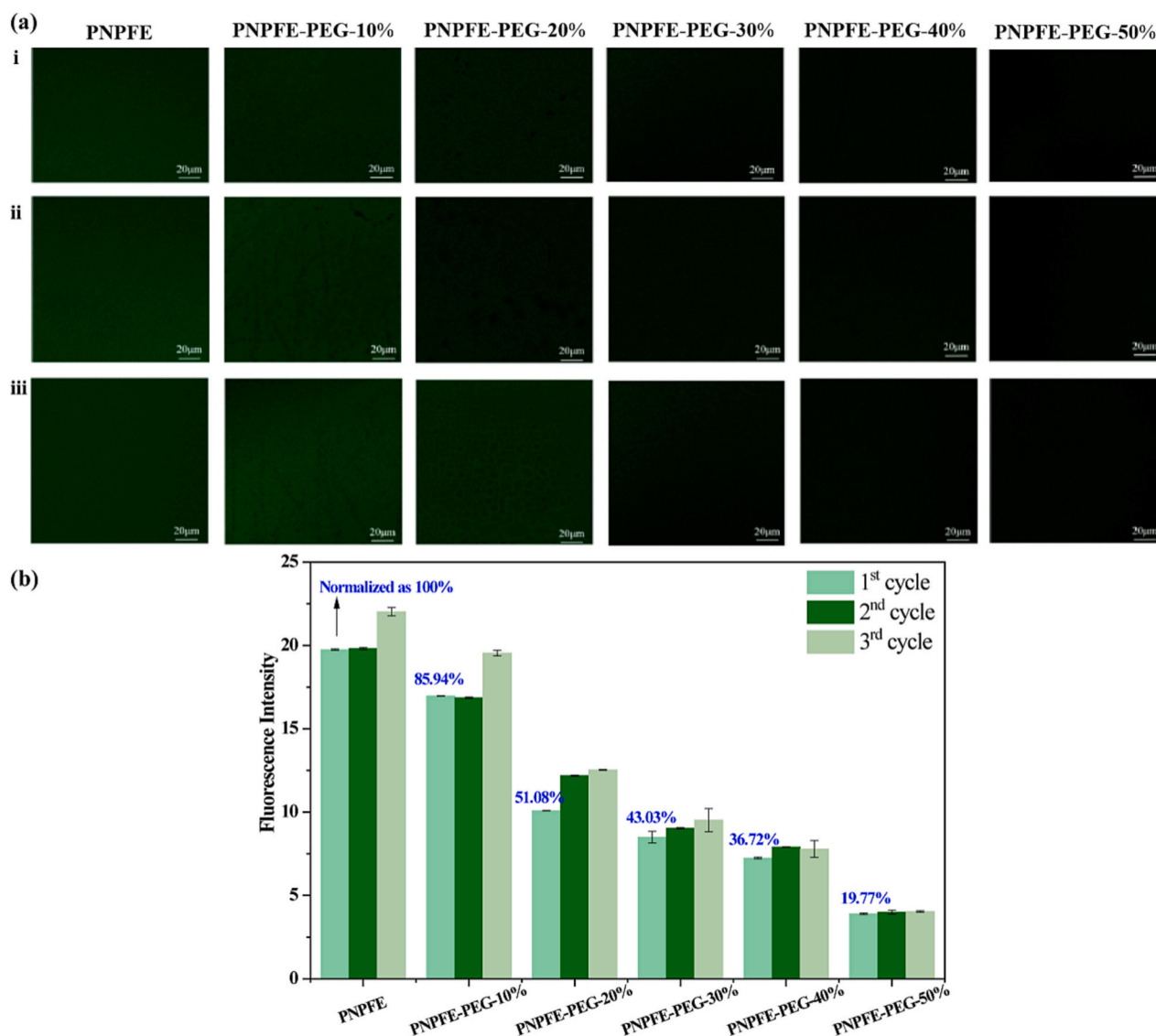


Fig. 5. CLSM image of three-cycle experiment of FITC-BSA adhered on the surface (a) PNPFE-PEG-x (i, ii and iii represent 1st cycle, 2nd cycle, 3rd cycle) and fluorescence intensity (b).

The UV spectrophotometer was utilized to quantitatively assess the protein resistance (anti-BSA) of the copolymer-coated surfaces, as depicted in Fig. 6. The initial protein adsorption on the fluorine-containing PNPFE surface of $53 \mu\text{g}/\text{cm}^2$ indicated the PNPFE polymer endowed materials protein release ability to some degree, attributed to its low surface energy and hydrophobicity. Increasing the NPEG content from 10 wt% to 50 wt% gradually decreased the amount of protein adsorption ($42\text{--}19 \mu\text{g}/\text{cm}^2$) on the amphiphilic polymer surfaces, exhibiting enhanced anti-fouling properties. Notably, PNPFE-PEG-50 % demonstrated exceptional anti-protein adhesion properties, as the protein adsorption capacity of the PNPFE-PEG-50 % surface was only one-third of that of the PNPFE surface, which is comparable/superior to the coumarin-based anti-BSA coating [40] and amphiphilic block copolymer PDMS-PDMAEMA anti-fouling coating [44]. Additionally, the adsorption amount of fibrinogen (Fg) protein on amphiphilic copolymer surfaces was determined. Likewise, the Fg protein adsorption ($67\text{--}27 \mu\text{g}/\text{cm}^2$) decreased gradually as the NPEG content increased from 0 to 50 wt%. Regarding the samples with the same NPEG content, the BSA protein adsorption capacity on the PNPFE-PEG-*x* surfaces was significantly lower than that of Fg protein, suggesting that PNPFE-PEG-*x* exhibited better BSA protein resistance. As a result, the PNPEG and PNPFE in PNPFE-PEG polymers collaborate to reduce the adhesion of protein and prevent biological deposition. Overall, these results demonstrated exceptional protein resistance and durability in anti-fouling performances through rational manipulation of molecular structures of PNPFE-PEG.

3.7. Anti-icing ability

In addition to protein resistance, it was observed that PNPFE-PEG-*x* displayed a degree of anti-icing behavior. The water freezing temperature on the PNPFE-PEG-*x* surfaces decreased gradually from -2°C to -8°C with an increase in NPEG content from 0 to 50 wt% (Fig. S9), as determined with a low-temperature ice shear strength testing device. Based on the above analysis, the hydrophilic NPEG segment on the surface strongly bind with water molecules, leading to the formation of non-frozen water for the surface hydrogen-bonded water molecules, thus delaying the formation of ice crystals. Besides, the hydrophobic PNPFE segments also play a positive role to prevent excessive water absorption. Moreover, the microphase separation of nanoscale surface topographies occurred on the coating surfaces, probably reducing ice adhesion strength and thus inhibiting ice formation in extreme environments [45–47].

4. Conclusion

We report norbornene-based amphiphilic copolymers bearing hydrophilic NPEG and hydrophobic fluorinated NPFE side chains with well-tailored molecular compositions through living ROMP. The molecular integrated design aimed to ensure simultaneous mechanical robustness and fouling resistance to meet the potential applications in air and aqueous environments. The molecular control over the mechanical properties and protein resistance has been systematically investigated. As design, hydrophobic NPFE segments imparts high strength and hydrophilic NPEG contributes to excellent chain mobility. Water molecules can significantly increase mechanical strength while restricts chain mobility through the formation of H-bonds with NPEG components. Consequently, the well-designed norbornene-based amphiphilic copolymers exhibited exceptional mechanical strength ($>8.73 \text{ MPa}$) in both water and air environments. As for the anti-protein ability, the protein adsorption capacity on the PNPFE surface of $53 \mu\text{g}/\text{cm}^2$ demonstrating a certain degree of anti-protein properties due to their low surface energy and hydrophobicity. The incorporation of NPEG can effectively reduce the protein adsorption and anti-fouling durability due to the microphase separation with nanoscale morphologies and surface reconstruction and formation of water layer upon

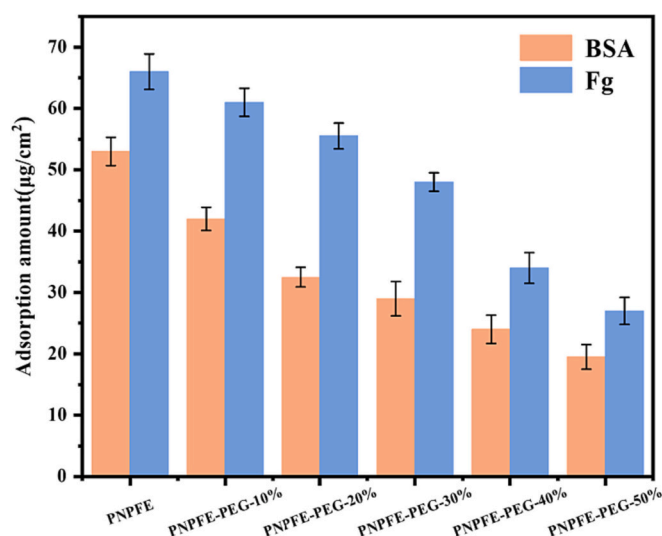


Fig. 6. Protein adsorption amount of the amphiphilic coatings (PNPFE-PEG-*x*).

immersion for inhibition protein adsorption. Additionally, the amphiphilic copolymers exhibited anti-icing property, with the freezing temperature of -2°C decreased to -8°C as the NPEG content increased. These results indicated the cooperative effect of molecular compositions on the properties. This study provides an effective strategy for the design of simultaneous enhanced mechanical and anti-fouling coatings and has a broad application prospect in different environments.

CRediT authorship contribution statement

Zhilu Gao: Writing – original draft, Methodology, Formal analysis. **Xuan Zhao:** Writing – original draft, Methodology, Investigation. **Guangzeng Luo:** Methodology, Formal analysis. **Wenqi Zhang:** Methodology. **Hang Wang:** Methodology, Investigation, Data curation. **Chuanyong Zong:** Writing – review & editing, Funding acquisition. **Lan Lei:** Writing – review & editing, Investigation, Funding acquisition, Data curation. **Hui Li:** Writing – review & editing, Validation, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Electronic Supplementary Information (ESI) available: Synthetic routes, Structural characterization, EDS characterization, GPC spectrum and molecular weight Mechanical properties, Microscope images, and

Anti-icing performance. Supplementary data to this article can be found online at doi:<https://doi.org/10.1016/j.porgcoat.2024.108361>.

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